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EXAMINER	
CASTRO, PAUL A	

ART UNIT	PAPER NUMBER
3662	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/298,208	Applicant(s) ITO et al.	
	Examiner PAUL A CASTRO	Art Unit 3662	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 02/05/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 13-23 is/are pending in the application.

5a) Of the above claim(s) 16-23 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 13-15 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☑ Claim(s) 16-23 are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 05/28/2021 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

This action is in response to the amendments filed on 02/05/2024.

Claims 1-12 are canceled.

Claims 13-23 are new.

Election/Restrictions

Newly submitted **claims 16-23** are directed to an invention that is independent or distinct from the invention originally claimed for the following reasons:

A: Claims 13-15 [Fig. 4, Fig. 7], are targeted towards comparing/selectively acquiring particular vehicle communication signals parameters for control criteria.

B: Claims 16-23 [Fig. 11], are targeted towards identifying size measurements comparisons for control parameters.

The species are independent or distinct because claims to the different species recites the mutually exclusive characteristics of such species. In addition, these species are not obvious variants of each other based on the current record.

The species are independent or distinct because of the following: **Species A** and **B** are directed to different types of control schemes. The inventions are independently distinct from the other because: Species A is directed to control parameters based on communication/networking signals while Species B is directed to a control parameters based on comparing dimensional size measurements. Identifying/comparing signal components to operate requires completely

different hardware, control schemes and sensors than determining dimensional measurements to operate. The systems of species A do not require identifying the dimensional measurements of Species B and Species B do not require the signal parameters of Species A.

Accordingly, the prior art applicable to Species A would not likely be applicable to species B. The prior art applicable to Species B would not likely be applicable to Species A.

Examiner's Note: *Newly submitted restricted claims include previous broad use language of "vehicle information". Although prior canceled claim-set included such language, Examiner wants to note that by BRI with conjunction with the original claim set, and in light of the specification, a reasonable interpretation was made that such language was in reference to wireless network information/communication with possible thresholds. The new claim set has redefined the vehicle information to an alternative of dimensional size measurements of other vehicles. If this was presented in the original claims, a restriction would have been appropriate. As such, Examiner understands the new claim sets as a different species not originally presented.*

Applicant is required under 35 U.S.C 121 to elect a single disclosed species for prosecution on the merits to which the claims shall be restricted if no generic claim is finally held to be allowable.

There is an examination and search burden for these patentably distinct species due to their mutually exclusive characteristics. The species require a different field of search (e.g., searching different classes/subclasses or electronic resources, or employing different search queries); and/or the prior art applicable to one species would not likely be applicable to another species; and/or the species are likely to raise different non-prior art issues under 35 U.S.C. 101 and/or 35 U.S.C. 112, first paragraph.

Since applicant has received an action on the merits for the originally presented invention, this invention has been constructively elected by original presentation for prosecution on the merits. Accordingly, claim 16-23 are withdrawn from consideration as being directed to a non-elected invention. See 37 CFR 1.142(b) and MPEP § 821.03.

To preserve a right to petition, the reply to this action must distinctly and specifically point out supposed errors in the restriction requirement. Otherwise, the election shall be treated as a final election without traverse. Traversal must be timely. Failure to timely traverse the requirement will result in the loss of right to petition under 37 CFR 1.144. If claims are subsequently added, applicant must indicate which of the subsequently added claims are readable upon the elected invention.

Should applicant traverse on the ground that the species are not patentably distinct, applicant should submit evidence or identify such evidence now of record showing the inventions to be obvious variants or clearly admit on the record that this is the case. In either instance, if the examiner finds one of the species unpatentable over the prior art, the evidence or admission may be used in a rejection under 35 U.S.C. 103 or pre-AIA 35 U.S.C. 103(a) of the other species.

Response to Arguments

1. Applicant's arguments with respect to claims have been considered but are moot because the arguments do not apply to any of the references being used in the current rejection.

112 Rejection

This rejection is withdrawn. New claims have been submitted and are currently examined.

101 Rejection

This rejection is withdrawn. New claims have been submitted and are currently examined.

103 Rejection

This rejection is withdrawn. New claims have been submitted and are currently examined.

Note: Applicant states that prior art Horn discloses a self-vehicle follow system that specifies a vehicle that is in-between a first vehicle and a third vehicle while the current invention specifies the use of the self-vehicle behind a first vehicle and a second vehicle and therefore the prior art does not disclose the system of the claimed invention. The Examiner disagrees by the following rationale.

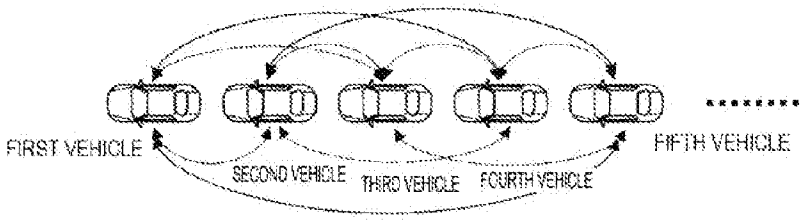
Prior art Horn does disclose a vehicle follow system with focus on the control features of the center vehicle however as stated by Horn, the focus on the control scheme for the center vehicle is more for simplifying the explanations for the invention, not that its system is specific to only function as a control scheme of the center vehicle **[see at least ¶ 78]**.

In other words, the control system is based on bilateral feedback using information that flows in both directions-from back to front and from front to back-and not just from front to back. In order to avoid ambiguity, we use the terms, "following vehicle," "controlled vehicle," and "leading vehicle" for three vehicles following one another as 'F', 'C', and 'L' do in FIG. 1.

The focus is on the control system of the center or "controlled" vehicle. Of course, the leading and following vehicles may also have such control systems, but to avoid ambiguity we focus on the control system of the center or "controlled" vehicle in the description of the invention.

Horn even specifies, "the vehicle controller system provides an output to the mechanism to control forward motion of the vehicle as a function of the outputs of the one or more leading distance sensors, of the one or more leading speed sensors, or the one or more following distance sensors…" **[¶ 27]**. Here it is understood that prior art Horn functions with intercommunication of all vehicles in communication with each other and pull from multiple vehicle in front and behind. Additionally, when looking at the Applicant's invention, Fig. 2 even discloses that platoon formation of vehicles communicate with each other both front and back and is understood that any position is capable of the same features.

Fig. 2



Prior art Horn is understood to disclose a system of platooning vehicles in communication with all leading vehicles as well as with all following vehicles.

Claim Objections

Claim 13 is objected to because of the following informalities: In line 23 wherein recited "information between said one vehicle and the third vehicle" is objected to for lack of proper antecedent basis. The claim is understood to refer to "one of the first vehicle and the second vehicle…".

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 13-15 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

In the art rejections below the claims have been treated as best understood by the examiner. Any claim not explicitly rejected under this heading is rejected as being dependent on an indefinite claim.

Wherein **claim 13** recites, "the third vehicle selectively receives vehicle information from one of the first vehicle and the second vehicle from which electrical power is received by the third vehicle in wireless communication" is unclear. How do these communication systems

receive electric power? Is electricity being shared between the vehicles? Are the individual vehicles receiving power from each other? Are they producing power? For one of ordinary skill in the art for smart vehicles, the possibilities of receiving electrical power for powering vehicle torque is possible through fairly complex systems, but due to a lack of details of in the claim language, it is not clear to what and how receiving electrical power is done specifically. Looking into the specification, signal strength is a consideration but stating signal strength being explicitly understood as receiving electrical power is not a clear way of being stated. Without explicit definitions or additional details to the claim language, it is not clear to what electric power received is referenced to. Examiner recommends adding clarifying language to how the electrical power received is processed and further defining what electrical power pertains to. To promote prosecution, the claim is understood use the best fit signals for reliable communication.

Claim 14 recites similar limitations and rejected by similar rationale of claim 13 above.

Examiner's Note

Examiner has cited particular paragraphs / columns and line numbers or figures in the references as applied to the claims below for the convenience of the applicant. Although the specified citations are representative of the teachings in the art and are applied to the specific limitations within the individual claim, other passages and figures may apply as well. It is respectfully requested from the applicant, in preparing the responses, to fully consider the references in entirety as potentially teaching all or part of the claimed invention, as well as the context of the passage as taught by the prior art or disclosed by the examiner. Applicant is reminded that the Examiner is entitled to give the broadest reasonable interpretation to the

language of the claims. Furthermore, the Examiner is not limited to Applicants' definition which is not specifically set forth in the claims.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 13-14 are rejected under 35 U.S.C. 103 as being unpatentable over US 20140236449 ("Horn").

As per claim 13 Horn a vehicle control apparatus mounted on a third vehicle in a vehicle following running system in which a first vehicle, a second vehicle subsequent to the first vehicle, and the third vehicle subsequent to the second vehicle carry out following running by being non- mechanically connected **[¶ 27 vehicle controller system provides an output to the mechanism to control forward motion…. As a function of the outputs of the one or more leading distance sensors, of the one or more leading speed sensors, ¶ 78 Of course, the**

leading and following vehicles can have such control systems, Fig. 1 (see also Response to arguments for 103 rejection above)],

the vehicle control apparatus being configured to:

selectively acquire first vehicle information including a first physical amount regarding a motion of the first vehicle and second vehicle information including a second physical amount regarding a motion of the second vehicle **[¶ 22 "adaptive cruise control" rely on information about a leading vehicle, ¶ 23 front to back controls… based on speed and distance (known in the art), ¶ 24 a center vehicle controls both front and following vehicles… all vehicles are provided with the control system ¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 1 vehicles F, C, and L, Fig. 4 (input) of leading distance and speed sensors (understood to come from more than one leading vehicle as explained above)**],

determine a third physical amount regarding a motion of the third vehicle based on the vehicle information acquired selectively from the first vehicle information and the second vehicle information**[¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands, ¶ 78 (all vehicles are**

understood to either first, second, or third vehicle), Fig. 4 forward motion control of vehicle], and

output an instruction based on the determined third physical amount to a braking apparatus or driving apparatus and a steering apparatus of the third vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands]**, wherein

based on the instruction, the driving apparatus applies a driving torque, the braking apparatus applies a hydraulic pressure, and a steering apparatus applies a steering angle, thereby controlling the braking or driving and steering of the third vehicle **[¶ 28 provide fuel and braking commands]**, and

in the selective acquisition of vehicle information, the third vehicle selectively receives vehicle information from one of the first vehicle and the second vehicle from which electrical power is received by the third vehicle in wireless communication of the vehicle information between said one vehicle and the third vehicle, the electric power being greater than or equal to a threshold value and being greater than electric power received from the other of the first and the second vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 109 Augments the information from sensors on the center vehicle with information from sensors in the leading and following vehicles… advantage of this approach is robustness due to redundancy, since the failure of a wireless link (predetermined condition, any failure of communication link, can use other communication links, such as those in the other vehicle instead)]**.

Horn describes the vehicle control system as a leading vehicle, center control vehicle, and a following vehicle. Horn further describes the control vehicle as handling sensor data of both the front leading vehicle and back following vehicle. Horn then states that the "control vehicle" label is for purposes of explaining the invention and avoid ambiguity and however, that all three vehicles possess the vehicle control system and are all capable of bi-direction communication for the control system **[¶ 78]**.

Horn describes different order of vehicles however; it would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to understand the vehicle control system of Horn would include accepting data from vehicles in any order for purposes of control automation of vehicle processions to minimize different acceleration jerkiness between vehicles to increase vehicle comfort and efficiency.

As per claim 14 Horn discloses a vehicle control apparatus mounted on a third vehicle in a vehicle following running system in which a first vehicle, a second vehicle subsequent to the first vehicle, and the third vehicle subsequent to the second vehicle carry out following running by being nonmechanically connected **[¶ 27 vehicle controller system provides an output to the mechanism to control forward motion…. As a function of the outputs of the one or more leading distance sensors, of the one or more leading speed sensors, ¶ 78 Of course, the leading and following vehicles can have such control systems, Fig. 1 (see also Response to arguments for 103 rejection above)]**,

the vehicle control apparatus being configured to:

selectively acquire first vehicle information including a first physical amount regarding a motion of the first vehicle and second vehicle information including a second physical amount regarding a motion of the second vehicle**[¶ 22 "adaptive cruise control" rely on information about a leading vehicle, ¶ 23 front to back controls… based on speed and distance (known in the art), ¶ 24 a center vehicle controls both front and following vehicles… all vehicles are provided with the control system ¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 1 vehicles F, C, and L, Fig. 4 (input) of leading distance and speed sensors (understood to come from more than one leading vehicle as explained above)]**,

determine a third physical amount regarding a motion of the third vehicle based on the vehicle information acquired selectively from the first vehicle information and the second vehicle information**[¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 4 forward motion control of vehicle]**,, and

output an instruction based on the determined third physical amount to

a braking apparatus or driving apparatus and a steering apparatus of the third vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands]**, wherein

based on the instruction, the driving apparatus applies a driving torque, the braking apparatus applies a hydraulic pressure, and a steering apparatus applies a steering angle, thereby controlling the braking or driving and steering of the third vehicle [**¶ 28 provide fuel and braking commands**, and

in the selective acquisition of vehicle information, the third vehicle selectively receives vehicle information from one of the first vehicle and the second vehicle from which electric power is received by the third vehicle in wireless communication between said one vehicle and the third vehicle, the electric power being greater than or equal to a threshold value and coming from said one vehicle ahead of the other **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 109 Augments the information from sensors on the center vehicle with information from sensors in the leading and following vehicles… advantage of this approach is robustness due to redundancy, since the failure of a wireless link (predetermined condition, any failure of communication link, can use other communication links, such as those in the other vehicle instead)]**.

Horn describes the vehicle control system as a leading vehicle, center control vehicle, and a following vehicle. Horn further describes the control vehicle as handling sensor data of both the front leading vehicle and back following vehicle. Horn then states that the "control vehicle"

label is for purposes of explaining the invention and avoid ambiguity and however, that all three vehicles possess the vehicle control system and are all capable of bi-direction communication for the control system **[¶ 78]**.

Horn describes different order of vehicles however; it would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to understand the vehicle control system of Horn would include accepting data from vehicles in any order for purposes of control automation of vehicle processions to minimize different acceleration jerkiness between vehicles to increase vehicle comfort and efficiency.

Claims 15 are rejected under 35 U.S.C. 103 as being unpatentable over US 20140236449 ("Horn") in further view of US 20120147802 ("Ukita").

As per claim 15 Horn discloses a vehicle control apparatus mounted on a third vehicle in a vehicle following running system in which a first vehicle, a second vehicle subsequent to the first vehicle, and the third vehicle subsequent to the second vehicle carry out following running by being nonmechanically connected **[¶ 27 vehicle controller system provides an output to the mechanism to control forward motion…. As a function of the outputs of the one or more leading distance sensors, of the one or more leading speed sensors, ¶ 78 Of course, the leading and following vehicles can have such control systems, Fig. 1 (see also Response to arguments for 103 rejection above)],**

the vehicle control apparatus being configured to:

selectively acquire first vehicle information including a first physical amount regarding a motion of the first vehicle and second vehicle information including a second physical amount regarding a motion of the second vehicle**[¶ 22 "adaptive cruise control" rely on information about a leading vehicle, ¶ 23 front to back controls… based on speed and distance (known in the art), ¶ 24 a center vehicle controls both front and following vehicles… all vehicles are provided with the control system ¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 1 vehicles F, C, and L, Fig. 4 (input) of leading distance and speed sensors (understood to come from more than one leading vehicle as explained above)]**,

determine a third physical amount regarding a motion of the third vehicle based on the vehicle information acquired selectively from the first vehicle information and the second vehicle information **[¶ 25 vehicle control system outputs sensor data (distance speed…), ¶ 26 vehicle controller system can be selectively masked, ¶ 27 vehicle control system accepts vehicle output data from other vehicles as inputs for control functions of vehicle, can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands, ¶ 78 (all vehicles are understood to either first, second, or third vehicle), Fig. 4 forward motion control of vehicle]**, and

output an instruction based on the determined third physical amount to a braking apparatus or driving apparatus and a steering apparatus of the third vehicle **[¶ 27 can choose which leading vehicle outputs to base self-vehicle controls on, ¶ 28 provide fuel and braking commands]**, wherein

based on the instruction, the driving apparatus applies a driving torque, the braking apparatus applies a hydraulic pressure, and a steering apparatus applies a steering angle, thereby controlling the braking or driving and steering of the third vehicle [**¶ 28 provide fuel and braking commands**], and

in the selective acquisition of vehicle information, the third vehicle selectively receives vehicle information from one of the first vehicle and the second vehicle, wherein the vehicle information has a time delay in wireless communication between said one vehicle and the third vehicle that is shorter than or equal to a threshold value **[¶ 108 sensors on the center vehicle are replaced by sensors in the leading… vehicles… a system using information conveyed from leading and following vehicles may benefit from the potentially higher accuracy and timeliness of the information** (*improved timeliness is understood as time delays in wireless communications is considered and having multiple sensor data links from different vehicles are available, with the advantage being having an option to choose a more timely data source*)**, ¶ 109 One advantage is robustness due to redundancy, since the failure of a wireless link does not disable the bilateral control… if combination of sensors… from leading… vehicles is used, then failure of an individual information source does not perturb the stability** (*failure of sensors is understood as a threshold to use one communication link over another*)] and that a higher timeliness is considered in wireless communication between the other of the first and the second vehicle and the third vehicle **[¶ 108 sensors on the center vehicle are replaced by sensors in the leading… vehicles… a system using information conveyed from leading and following vehicles may benefit from the potentially higher accuracy and timeliness of the information** (*improved*

timeliness is understood as time delays in wireless communications is considered and having multiple sensor data links from different vehicles are available, advantage is having an option to choose a more timely data sources)**].**

Horn describes the vehicle control system as a leading vehicle, center control vehicle, and a following vehicle. Horn further describes the control vehicle as handling sensor data of both the front leading vehicle and back following vehicle. Horn then states that the "control vehicle" label is for purposes of explaining the invention and avoid ambiguity and however, that all three vehicles possess the vehicle control system and are all capable of bi-direction communication for the control system **[¶ 78]**.

Horn describes different order of vehicles however; it would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to understand the vehicle control system of Horn would include accepting data from vehicles in any order for purposes of control automation of vehicle processions to minimize different acceleration jerkiness between vehicles to increase vehicle comfort and efficiency.

Horn is not explicit to using the shorter time delay in wireless communication.

Ukita discloses a threshold time delay **[¶ 47 the calculated transmission period is longer than the shortest allowable delay time in the allowable delay times]** and using the shorter time delay in wireless communication **[¶ 47 the communication device determines the shortest allowable delay time as the transmission period]**.

It would have been obvious to one of ordinary skill in the art before the effective filing date the invention was made to modify Horn with the teachings of Ukita to use the most efficient wireless data pathway to increase data transmission efficiency by determining fastest transmissions for the most up to date data information transfer for improved critical data updates.

Additional Art to Consider

Application **Pub. No. 20170168503** titled *AUTONOMOUS VEHICLE TOWING SYSTEM AND METHOD,* further includes a system inter-vehicle communications used for autonomous vehicle follow systems. **This is similar to the Applicant's invention in that Applicant's** in using wireless non-mechanical connections to commutate between vehicles for a leading/following relation between said vehicles.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to PAUL A CASTRO whose telephone number is (571)272-4836. The examiner can normally be reached 10-6pm on campus.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jelani Smith can be reached on 5712703969. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available

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PAUL A. CASTRO
Examiner
Art Unit 3662

/P.A.C/
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/JELANI A SMITH/
Supervisory Patent Examiner, Art Unit 3662